

Industrial Oriented Mini Project (IOMP)

on

AI Attendance System

Submitted in partial fulfilment of the requirement for the award of degree

Bachelor of Technology

in

Computer Science and Engineering

by

ANGULURI HARI CHARAN- 23J21A0506

DANAPALLI SRINIVAS- 23J21A05032

EGYARAPU MADHAN- 23J21A0539

GADDI KOPULA VENKATESH- 23J21A0544

Under the Supervision

of

Mr.Ch.China Subba Reddy M.Tech
Assistant Professor



Department of Computer Science and Engineering

JOGINPALLY B.R. ENGINEERING COLLEGE
(UGC AUTONOMOUS)

Accredited by NAAC with 'A+' Grade, Recognized under Sec. 2(f) of UGC Act. 1956
Approved by AICTE, Affiliated to JNTUH, Hyderabad and ISO 9001:2015 Certified
Bhaskar Nagar, Yenkapally, Moinabad (Mandal)
R.R. (Dist)-500075. T.S., India

ACADEMIC YEAR 2025-26

JOGINPALLY B.R. ENGINEERING COLLEGE
(UGC AUTONOMOUS)

Accredited by NAAC with 'A+' Grade, Recognized under Sec. 2(f) of UGC Act. 1956
Approved by AICTE, Affiliated to JNTUH, Hyderabad and ISO 9001:2015 Certified
Bhaskar Nagar, Yenkapally, Moinabad (Mandal)
R.R (Dist) - 500075. T.S., India



CERTIFICATE

The seminar entitled **AI ATTENDANCE SYSTEM** that is been submitted by **ANGULURI HARI CHARAN-23J21A0506, DANAPALLI SRINIVAS- 23J21A05032, EGYARAPU MADHAN-23J21A0539, GADDI KOPULA VENKATESH- 23J21A0544**, in partial fulfilment of the award of Bachelor of Technology in Computer Science and Engineering to Jawaharlal Nehru Technological University Hyderabad. It is record of bonafide work carried out under our guidance and supervision. In my opinion, this report is of standard required for the degree of Bachelor of Technology.

PROJECT SUPERVISOR

Mr.Ch.China Subba Reddy M.Tech,
Assistant professor

HEAD OF THE DEPARTMENT

Dr. T. Prabakaran M.E., Ph.D.,
Professor

EXTERNAL EXAMINER

DECLARATION

We hereby declare that the Project entitled “**AI ATTENDANCE**”, submitted towards the completion of a Industrial Oriented Mini Project Work in the III-II B.Tech., CSE at Joginpally B.R. Engineering College, Hyderabad, is an authentic record of our original work.

ANGULURI HARI CHARAN- 23J21A0506

DANAPALLI SRINIVAS- 23J21A05032

EGYARAPU MADHAN- 23J21A0539

GADDI KOPULA VENKATESH- 23J21A0544

ACKNOWLEDGMENT

We would like to express our sincere thanks to **Dr N. Seetharamaiah, Principal of Joginpally B.R. Engineering College**, Hyderabad, for his support by providing the working facilities in the college.

Our sincere thanks and gratitude to **Dr.T.Prabakaran, Professor & HOD, Department of Computer Science and Engineering, Joginpally B.R. Engineering College**, Hyderabad, for all timely support and valuable suggestions during the period of our project.

We express gratitude to our Project Coordinator **Mr.C.Sharathchandra, Assistant Professor, Department of Computer Science and Engineering, Joginpally B.R. Engineering College**, Hyderabad, for his valuable suggestions and advices.

We are extremely thankful to our Project Supervisor, **Mr.Ch.China Subba Reddy, Assistant professor, Department of Computer Science and Engineering, Joginpally B.R. Engineering College**, Hyderabad, for his constant guidance and encouragement throughout the project.

Finally, we would like to thank all Faculty and Staff of CSE department who helped us directly or indirectly. We wish to acknowledge our Parents and Friends for giving moral strength and constant encouragement.

ANGULURI HARI CHARAN- 23J21A0506

DANAPALLI SRINIVAS- 23J21A05032

EGYARAPU MADHAN- 23J21A0539

GADDI KOPULA VENKATESH- 23J21A0544

ABSTRACT

The AI Face Attendance System is an advanced intelligent automation solution developed to overcome the limitations of traditional attendance methods such as manual registers, ID cards, and biometric systems, by introducing a fully contactless, efficient, and highly accurate approach based on face recognition technology. This system utilizes the power of computer vision and machine learning algorithms to capture, detect, and recognize human faces in real time using a webcam, thereby enabling automatic attendance marking without requiring any physical interaction or manual input. By continuously analysing video frames, the system identifies unique facial features of individuals and compares them with a pre-trained dataset to verify identity with high precision. The implementation of this project is carried out using Python, supported by robust libraries such as OpenCV for image processing and face detection, NumPy for handling numerical computations, and Pandas for organizing and managing attendance data in structured formats like CSV or Excel files. The system follows a well-defined pipeline that includes image acquisition, face detection, dataset creation, model training using algorithms such as LBPH, and real-time face recognition, culminating in automated attendance recording with timestamps. This approach significantly minimizes human intervention, eliminates the possibility of proxy attendance, reduces operational errors, and enhances overall efficiency and reliability. Furthermore, the system is scalable and adaptable to various environments such as educational institutions, corporate offices, and secure facilities, making it a practical and modern solution for attendance management. By integrating automation with intelligent recognition capabilities, the AI Face Attendance System not only streamlines the attendance process but also contributes to improved data accuracy, security, and time management in organizational workflows.

S.NO.	List of Figures / Block Diagrams	Page No.
1	Face Recognition Attendance	1
2	Scope	3
3	Camera Module	5
4	Face Recognition	6
5	Face Recognition in Project	6
6	Data Flow Diagram	10
7	Face Detection	12
8	Face Detection Module	13
9	Model Training Implementation	14
10	Face Recognition Implementation	15
11	Module Implementation	16
12	Registers new users	17
13	Captures face images	18
14	Trains the Model	19
15	Recognizes faces in real time	20
16	Marks attendance automatically	21
17	Registration Output	22
18	Recognition Output	23
19	User Interface Menu	37
20	Recognition Output	40

CHAPTER	TABLE OF CONTENTS	Page No.
	Abstract	i
	List of Figures / Block Diagrams	ii
1	INTRODUCTION	1
	1.1 Overview	1
	1.2 Objectives	2
	1.3 Scope	3
2	EXISTING SYSTEM	4
3	PROPOSED SYSTEM	5
4	TECHNOLOGIES USED	6
	4.1 Python	6
	4.2 Open CV	6
	4.3 NumPy	6
	4.4 Pandas	6
	4.5 CSV	6
5	SYSTEM ARCHITECHTURE	7
	5.1 Components	7
	5.1.1 Camera Module:	7
	5.1.2 Face Detection Module:	7
	5.1.3 Face Training Module:	8
	5.1.4 Face Recognition Module:	8
	5.1.5 Attendance Storage Module:	9
	5.1.5 Attendance Storage Module:	9

CHAPTER	TABLE OF CONTENTS	Page No.
	5.2 Architecture Explanation	9
	5.3 Workflow	9
	5.3.1 Register New User:	9
	5.3.3 Train Model:	9
	5.3.4 Recognize Faces:	10
	5.3.5 Mark Attendance:	10
6	SYSTEM DESIGN & IMPLEMENTATION	11
	6.1 System Design	11
	6.2 Data Flow	12
	6.2.1 User Input	12
	6.2.2. Image Capture (Webcam)	12
	6.2.3. Face Detection (Haar Cascade)	12
	6.2.4. Face Extraction (ROI – Region of Interest)	13
	6.2.5. Feature Extraction (LBPH – Local Binary Patterns Histogram)	13
	6.2.6. Face Matching	13
	6.2.7. Attendance Marking	13
	6.2.8. CSV Storage	13
	6.3 Module Implementation	14
	6.3.1 Face Detection Implementation	14
	6.3.2 Registration Module Implementation	15
	6.3.3 Model Training Implementation	16

CHAPTER	TABLE OF CONTENTS	Page No.
	6.3.4 Face Recognition Implementation	17
	6.3.5 Attendance Module Implementation	16
7	RESULTS AND DISCUSSION	19
	7.1 System Output	19
	7.1.1 Registers new users	19
	7.1.2 Captures face images	20
	7.1.3. Trains the Model	21
	7.1.4. Recognizes faces in real time	22
	7.1.5. Marks attendance automatical	23
	7.2 Sample Output	24
	7.2.1 Registration Output	24
	7.2.2 Recognition Output	25
	7.2.3 Attendance Output (CSV)	25
8	SOURCE CODE IMPLEMENTATION	26
	8.1 Dataset Management	26
	8.2 Model Storage	26
	8.3. Face Detection Module	26
	8.4. Model Training Module	27
	8.5. Attendance Management Module	27
	8.6. User Registration Module	29
	8.7. Training Execution Module	31

CHAPTER	TABLE OF CONTENTS	Page No.
	8.8. Face Recognition Module	33
	8.9. Main System Controller	36
	8.10. Attendance Data Storage	37
	8.11. Student Information Storage	37
9	APPLICATIONS	38
	9.1 Educational Institutions	38
	9.2 Corporate Offices	38
	9.3 Government Organizations	38
	9.4 Events and Conferences	38
10	FINAL OUTPUT	39
	10.1 User Interface Menu	39
	10.1.1. Register Student	39
	10.1.2. Train Model	39
	10.1.3. Start Attendance	39
	10.1.4. Exit	39
	10.2 Recognition Output	40
11	CONCLUSION	41
12	REFERENCE	42

CHAPTER-I

INTRODUCTION

1.1 Overview

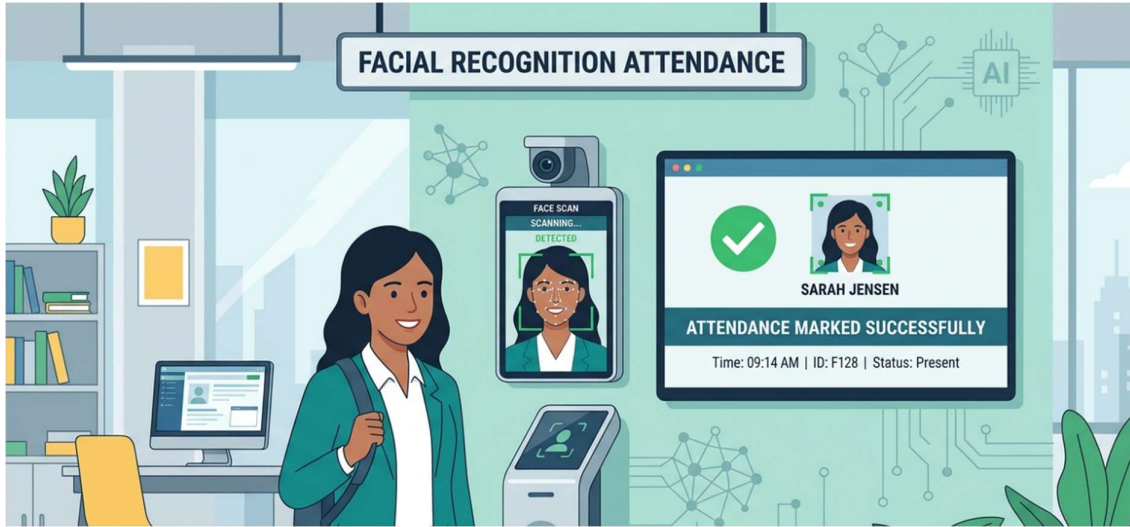


Figure 1:Face Recognition Attendance

Attendance tracking is a fundamental and routine activity in educational institutions and professional environments, where maintaining accurate records of presence is crucial for administration, evaluation, and accountability. Traditional methods such as manual registers, roll calls, and ID card-based systems have been widely used for decades; however, these approaches suffer from several limitations including inefficiency, time consumption, human errors, and susceptibility to manipulation such as proxy attendance. These drawbacks often lead to inaccurate data, increased administrative workload, and reduced overall reliability of attendance management systems.

With the rapid advancements in artificial intelligence and computer vision technologies, face recognition has emerged as a highly effective and reliable solution for automated identity verification. By analyzing unique facial features, modern systems can accurately distinguish individuals without requiring physical contact or manual intervention. The proposed AI-based face attendance system leverages these technological advancements to provide a smart, automated alternative to conventional attendance methods. It captures facial images through a webcam, processes them in real time, and matches them against a pre-trained dataset to identify individuals. Once a match is confirmed, the system automatically records attendance along with relevant details such as date and time.

This approach not only enhances efficiency by significantly reducing the time required for attendance marking but also improves security by preventing unauthorized or proxy entries. Furthermore, the system ensures higher data accuracy and consistency, making it a dependable solution for modern institutions. By integrating automation with intelligent recognition capabilities, the proposed system transforms traditional attendance tracking into a seamless, reliable, and technologically advanced process.

1.2 Objectives

The primary objective of this project is to design and implement an automated attendance system that utilizes face recognition technology to identify individuals based on their unique facial features. By integrating computer vision techniques, the system replaces traditional attendance methods with a smart, contactless solution that operates in real time, ensuring seamless and modern attendance management. Another important goal of the system is to minimize the need for manual intervention in the attendance process. Conventional methods such as roll calls or manual entry require significant time and effort, especially in large groups. The proposed system automates the entire process, allowing attendance to be recorded instantly, thereby saving valuable time and increasing overall efficiency.

The system is designed to enhance the accuracy of attendance records by using biometric facial data, which is unique to each individual. This prevents fraudulent practices such as proxy attendance, where one person marks attendance on behalf of another. By ensuring reliable identification, the system maintains integrity and trust in attendance tracking. A key objective is to maintain attendance data in a digital format, making it easy to store, access, and manage. The system records details such as name, date, and time in structured files like CSV or Excel, which can be easily analyzed or integrated with other systems. This reduces paperwork and ensures secure and organized data management.

The system is developed with scalability in mind, allowing it to be implemented across different environments such as schools, colleges, offices, and organizations of varying sizes. Its efficient design ensures consistent performance even with a large number of users, making it a flexible and future-ready solution for attendance management. By ensuring reliable identification, the system maintains integrity and trust in attendance tracking.

1.3 Scope



Figure 2:Scope

The scope of the AI Face Attendance System extends to various sectors where efficient and reliable attendance tracking is essential. In educational institutions, the system can be used to automatically record student attendance during classes, reducing the burden on teachers and ensuring accurate records. In offices and corporate environments, it helps monitor employee attendance, improve workforce management, and eliminate time theft or proxy entries. The system is also highly useful in government organizations, where maintaining secure and transparent attendance records is critical for administrative efficiency. Additionally, it can be applied in event management systems, where it enables quick and automated participant verification and attendance tracking during conferences, seminars, and large gatherings. Overall, the system offers a flexible and scalable solution that can be adapted to different domains requiring automated identity verification and attendance management.

CHAPTER-II

EXISTING SYSTEM

The existing attendance system is mainly based on manual methods such as maintaining paper registers or simple digital entry systems. In many institutions and organizations, attendance is recorded by calling out names or signing attendance sheets. Some organizations also use biometric systems like fingerprint scanners for attendance tracking.

However, these systems have several limitations. Manual attendance is time-consuming and prone to human errors such as incorrect entries or missed records. It is also easy to manipulate, allowing proxy attendance where one person marks attendance for another. Biometric systems, although more advanced, still require physical contact and can sometimes fail due to fingerprint issues or hardware problems. Additionally, managing and storing large amounts of attendance data becomes difficult and inefficient in these traditional approaches.

Biometric systems, while more advanced than manual methods, also have their own limitations. They require physical contact, which can be inconvenient and unhygienic, especially in shared environments. Fingerprint scanners may fail due to issues like worn-out fingerprints, dirt, or hardware malfunctions. Similarly, RFID-based systems can be misused if cards are shared among users. These systems also involve additional hardware costs and maintenance.

CHAPTER-III

PROPOSED SYSTEM

The proposed system is an AI-based Face Attendance System designed to automate the attendance process using face recognition technology. Unlike traditional methods, this system uses a webcam to capture images of individuals and identifies them based on their facial features. The recognition is performed using the LBPH (Local Binary Pattern Histogram) algorithm, which is efficient and suitable for real-time applications. Once a face is detected and recognized, the system automatically records attendance along with the person's name, date, and time in a CSV file.

The system is structured into several modules to ensure smooth functioning. These include user registration, where images of individuals are captured and stored; a training module, where the system learns facial patterns; a face detection module, which identifies faces in real-time; a recognition module, which matches detected faces with the trained data; and an attendance module, which records and stores the attendance data. This modular design makes the system organized, scalable, and easy to maintain.

One of the key advantages of the proposed system is that it operates in real time, allowing instant detection and marking of attendance without any delay. It eliminates the need for manual intervention and reduces the workload on staff. Since the system is based on facial recognition, it prevents proxy attendance, ensuring higher accuracy and reliability. Additionally, it is a contactless system, making it more hygienic and convenient compared to biometric methods.

The proposed system also improves data management significantly. Attendance records are stored digitally in an organized format, making it easy to access, update, and analyze the data. Reports can be generated quickly, and attendance trends can be monitored efficiently. The system can also be further enhanced by integrating with databases or cloud storage for large-scale applications.

CHAPTER-IV

TECHNOLOGIES USED

4.1 Python

Python is used as the primary programming language for developing the AI Face Attendance System due to its simplicity, readability, and extensive support for libraries and frameworks related to artificial intelligence, machine learning, and computer vision. Its easy-to-understand syntax allows for rapid development and implementation of complex functionalities such as face detection, recognition, and data processing. Python also provides strong community support and a wide range of pre-built modules, which significantly reduces development time and effort. Additionally, its compatibility with libraries like OpenCV, NumPy, and Pandas makes it an ideal choice for building a complete and efficient attendance system.

4.2 Open CV

OpenCV (Open Source Computer Vision Library) is used in this system for image processing and face detection tasks.

4.3 NumPy

NumPy is used in this system for performing numerical operations and efficiently handling arrays and matrices, which are fundamental in image processing and machine learning tasks.

4.4 Pandas

Pandas is used in this system for managing and organizing attendance records in a structured and efficient manner

4.5 CSV File

CSV (Comma-Separated Values) files are used in this system as a lightweight and simple database for storing student details and attendance records.

CHAPTER-V

SYSTEM ARCHITECHTURE

5.1 Components

5.1.1 Camera Module:

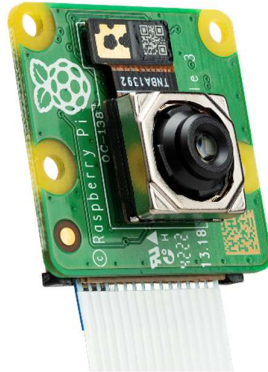


Figure 3: Camera Module

The Camera Module serves as the primary input unit of the system, responsible for capturing live video through a webcam or external camera device. It continuously streams frames at a specific frame rate, ensuring smooth and real-time data flow for further processing. This module may also perform basic preprocessing tasks such as adjusting brightness, contrast, and resolution to improve image quality under varying lighting conditions. In addition, it ensures proper alignment and positioning of faces within the frame, which is essential for accurate detection and recognition. A stable and high-quality camera feed directly impacts the overall performance and reliability of the system.

5.1.2 Face Detection Module:

The Face Detection Module is responsible for identifying the presence and location of human faces within each frame captured by the camera. It uses computer vision techniques such as Haar Cascade classifiers or more advanced methods like HOG (Histogram of Oriented Gradients) to scan the image and detect facial regions. This module filters out irrelevant background information and focuses only on the face area, which is then passed to the next stage. It can handle multiple faces in a single frame and may include optimizations like scaling and region-of-interest (ROI) selection to improve detection speed and accuracy. Efficient face detection is critical as it directly affects recognition performance.

5.1.3 Face Training Module:

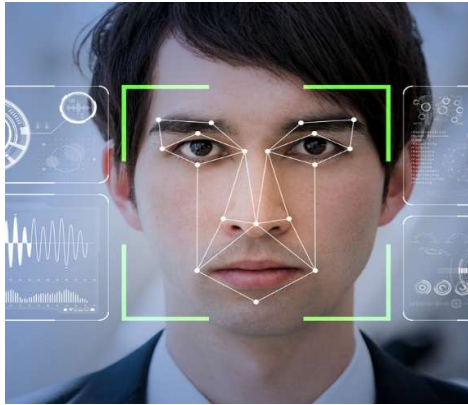


Figure 4: Face Recognition

The Face Training Module is responsible for building the recognition model by learning from a dataset of labeled facial images. It processes multiple images of each individual under different conditions such as lighting, angles, and expressions to improve robustness. Using algorithms like Local Binary Patterns Histogram (LBPH), Eigenfaces, or Fisherfaces, the system extracts unique facial features and converts them into numerical representations. These features are then used to train a model, which is saved (e.g., as a .yml file) for later use. A well-trained model ensures higher recognition accuracy and better performance in real-world scenarios.

5.1.4 Face Recognition Module:

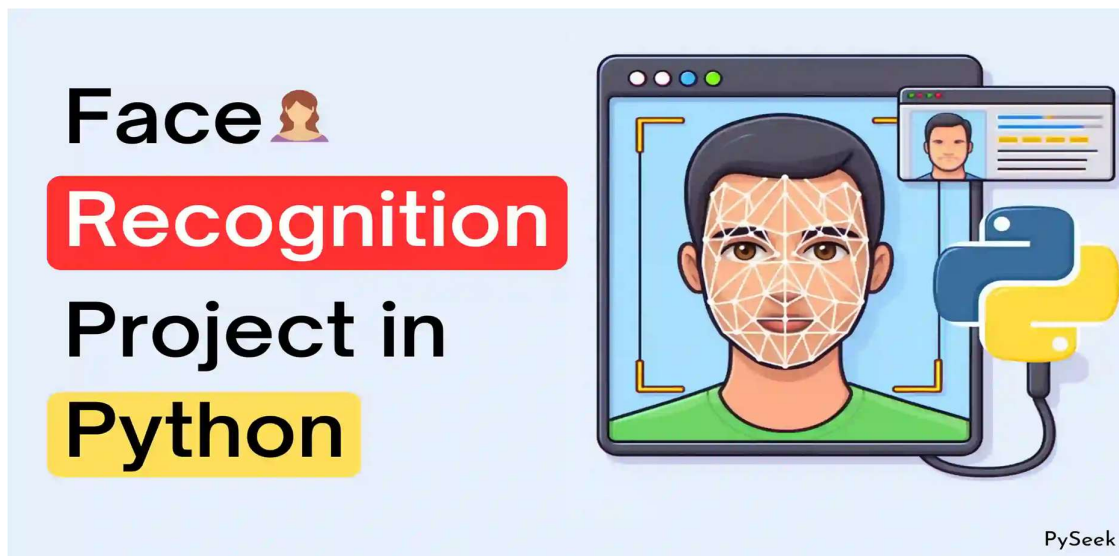


Figure 5: Face Recognition Module

The Face Recognition Module performs the core task of identifying individuals by comparing detected faces from live video with the trained model. It extracts features from the detected face and matches them against stored features in the database. Based on similarity scores or confidence levels, the system determines the identity of the person. If the confidence level meets a predefined threshold, the person is recognized; otherwise, they may be marked as unknown. This module operates in real time and must be optimized for speed and accuracy. It plays a key role in ensuring that attendance is recorded only for valid and recognized individuals.

5.1.5 Attendance Storage Module:

The Attendance Storage Module manages the recording and maintenance of attendance data in an organized and structured format.

5.2 Architecture Explanation

The architecture of the AI Face Attendance System is designed as a sequential pipeline that ensures smooth data flow from image capture to attendance recording. Initially, the system captures real-time images or video frames using a webcam, which serves as the primary input source.

5.3 Workflow

5.3.1 Register New User:

The workflow begins with the registration of a new user, which is a crucial step for building the system's database. During registration, the user provides basic details such as name, ID, department, or class, which are stored in the system.

5.3.2 Capture Face Images:

After registration, the system captures multiple images of the user's face using a webcam. Typically, several samples (e.g., 20–50 images) are taken to ensure better accuracy.

5.3.3 Train Model:

In the training phase, the system processes all the collected images to build a recognition model. It reads the dataset, extracts important facial features, and converts them into numerical representations.

5.3.4 Recognize Faces:

Once the model is trained, the system enters the recognition phase. It captures live video from the webcam and detects faces in each frame. The detected faces are then compared with the trained model to identify individuals. The system calculates a confidence score to determine how closely the detected face matches stored data.

5.3.5 Mark Attendance:

After successfully recognizing a face, the system proceeds to mark attendance automatically.

CHAPTER-VI

SYSTEM DESIGN & IMPLEMENTATION

6.1 System Design

The AI Face Attendance System is architected using a modular and layered design approach that emphasizes scalability, maintainability, efficiency, and real-time processing capabilities, making it suitable for academic institutions, offices, and secure environments where automated attendance tracking is required. At its core, the system is divided into multiple independent yet interconnected modules, each responsible for a specific stage of the overall workflow, including face detection, dataset generation, data preprocessing, model training, face recognition, attendance marking, and data storage management. This modular decomposition allows developers to isolate functionalities, making debugging easier, enhancing system reliability, and enabling future upgrades without affecting the entire system. The system operates in a pipeline architecture where data flows sequentially from one module to another, ensuring structured execution and minimizing redundancy.

Initially, the face detection module uses computer vision techniques to identify human faces from live video streams or captured images, typically leveraging pre-trained classifiers such as Haar Cascade or deep learning-based detectors, ensuring accurate and fast detection even under varying lighting conditions and backgrounds. Once a face is detected, the dataset creation module is triggered, which captures multiple images of the detected face from different angles and expressions to build a robust dataset; this dataset is then labeled with a unique user ID and stored systematically in directories, forming the foundation for model training. The preprocessing stage ensures that all images are normalized, resized, and converted into a consistent format to improve training efficiency and accuracy. Following this, the model training module utilizes machine learning algorithms such as Local Binary Patterns Histogram (LBPH), Eigenfaces, or Fisherfaces to extract facial features and create a trained model that can distinguish between different individuals; the trained model is saved in a structured format (e.g., YAML file), enabling reuse without retraining every time.

6.2 Data Flow

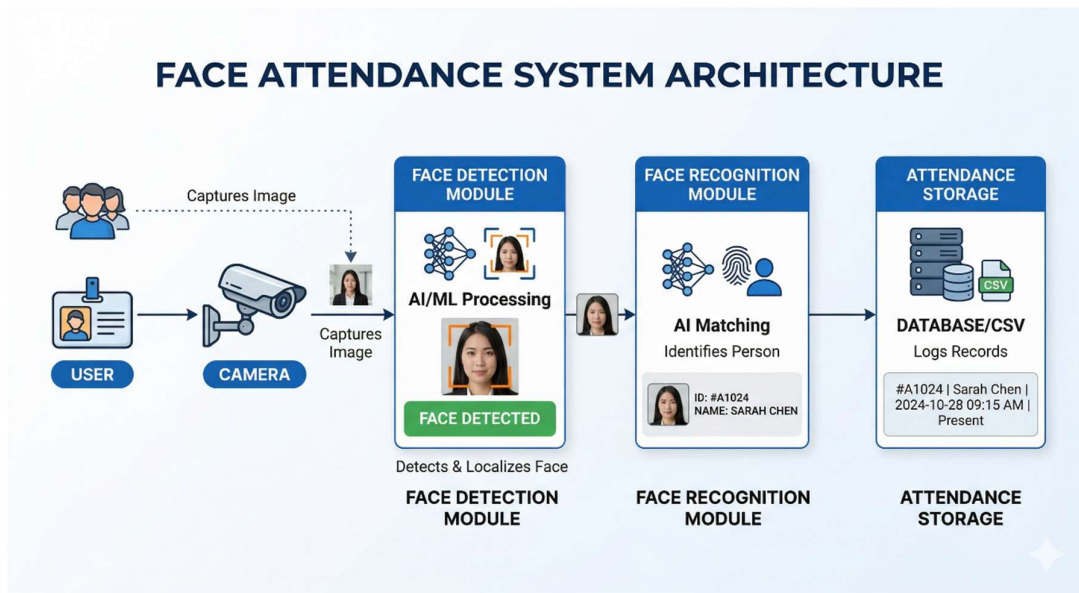


Figure 6:Data Flow Diagram

6.2.1 User Input

This is the initial step where the user interacts with the system. The user may start the application, register their face, or initiate attendance marking. The system takes basic input such as user ID, name (during registration), or simply detects presence (during attendance mode).

6.2.2. Image Capture (Webcam)

This module is responsible for acquiring real-time visual data using a webcam. The system continuously captures frames from the video stream using libraries like OpenCV. Each frame is processed individually to detect faces. To optimize performance, frames are often resized and converted into grayscale, as grayscale images reduce computational complexity while retaining essential features.

6.2.3. Face Detection (Haar Cascade)

Face detection is performed using the Haar Cascade Classifier, a machine learning-based approach trained on thousands of positive (face) and negative (non-face) images. It uses Haar-like features, which are patterns of intensity differences (such as edges and lines), to detect facial structures.

6.2.4. Face Extraction (ROI – Region of Interest)

Once a face is detected, the system extracts the Region of Interest (ROI), which is the portion of the image containing the face. This step is crucial because it isolates relevant data and removes background noise, improving recognition accuracy..

6.2.5. Feature Extraction (LBPH – Local Binary Patterns Histogram)

Feature extraction is one of the most critical stages, where the system converts raw face images into meaningful numerical representations. The LBPH algorithm works by analyzing local textures in the image.

6.2.6. Face Matching

In this stage, the extracted feature vector is compared with stored feature vectors in the trained model. The system calculates a similarity score (or distance metric) between the input face and known faces.

6.2.7. Attendance Marking

Once a face is successfully recognized, the system proceeds to mark attendance. It records details such as user ID, name, date, and time of recognition. To maintain accuracy, the system checks whether the attendance for that user has already been recorded for the current session or day, preventing duplicate entries.

6.2.8. CSV Storage

The final step involves storing attendance records in a CSV file. CSV (Comma-Separated Values) format is widely used due to its simplicity and compatibility with spreadsheet applications like Excel. Each record is stored as a row containing fields such as ID, name, date, and timestamp. The system ensures proper file handling, including creating new files if they do not exist and appending new records without overwriting existing data. Data validation techniques are applied to maintain consistency and avoid corruption. Although CSV is suitable for small to medium-scale systems, it can be replaced with databases like MySQL or MongoDB for large-scale deployments requiring faster querying and better data management.

6.3 Module Implementation

6.3.1 Face Detection Implementation

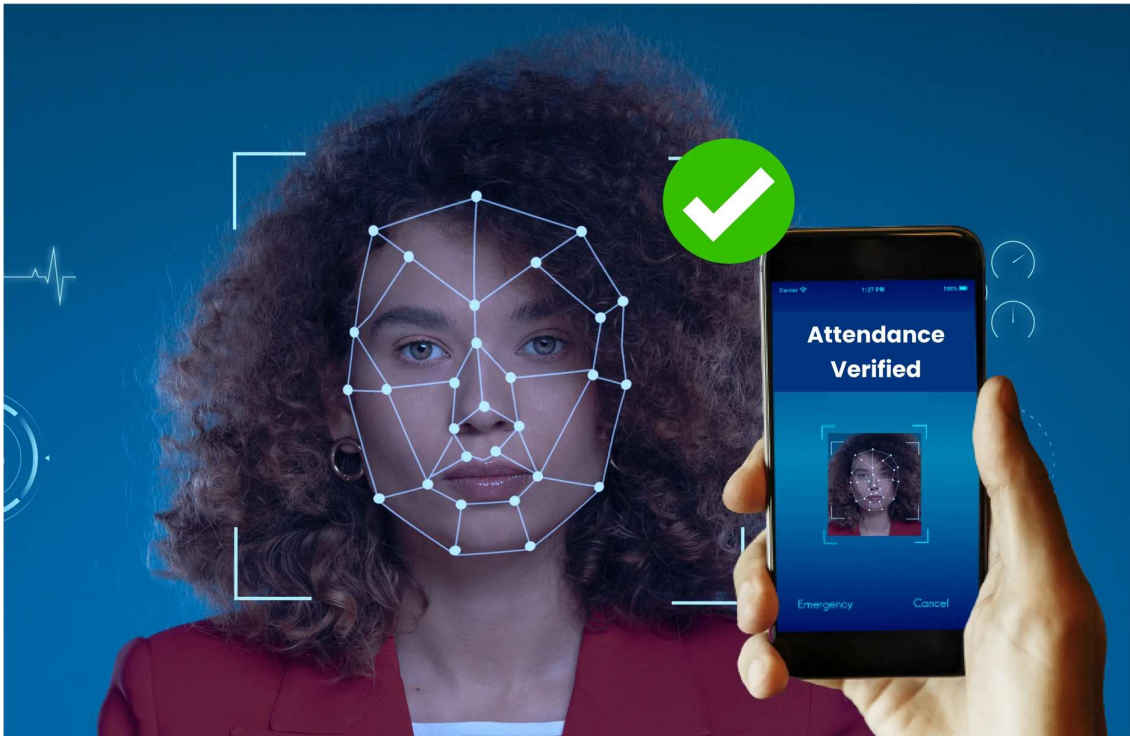


Figure 7:Face Detection

The Face Detection Implementation module in the AI Face Attendance System is a fundamental component responsible for accurately identifying human faces within input images or real-time video streams, and it is implemented using the widely adopted OpenCV library with the Haar Cascade classifier, which is a machine learning-based object detection approach known for its efficiency and suitability in real-time applications; the process begins by capturing frames from a webcam or loading input images, after which each frame is preprocessed to improve detection accuracy and reduce computational complexity, where one of the primary preprocessing steps involves converting the colored image (RGB format) into grayscale because grayscale images contain only intensity information and eliminate unnecessary color data, thereby significantly reducing the number of computations required and improving processing speed without affecting the detection capability, in addition to grayscale conversion, optional preprocessing techniques such as histogram equalization may be applied to enhance contrast, especially under poor lighting conditions.

6.3.2 Registration Module Implementation



Figure 8: Face Detection Module

The Registration Module Implementation in the AI Face Attendance System plays a crucial role in building the foundational dataset required for accurate face recognition and is responsible for capturing and storing facial images of users along with their corresponding identity information in a structured manner.

This module begins by collecting basic user details such as name and roll number through an input interface which may be command line based or graphical depending on the system design. These details are validated to ensure uniqueness and correctness so that no duplicate entries are created in the system. Once the user information is successfully recorded the module activates the webcam to start capturing facial images in real time. The camera continuously streams frames and the system processes each frame to detect the presence of a face using the face detection module. When a face is detected the system extracts the facial region and captures it as an image sample. This process is repeated multiple times to collect a sufficient number of face samples from different angles and expressions which improves the robustness and accuracy of the recognition model. Typically a predefined number of images such as 50 to 100 samples are captured for each user to ensure variability in the dataset.

6.3.3 Model Training Implementation

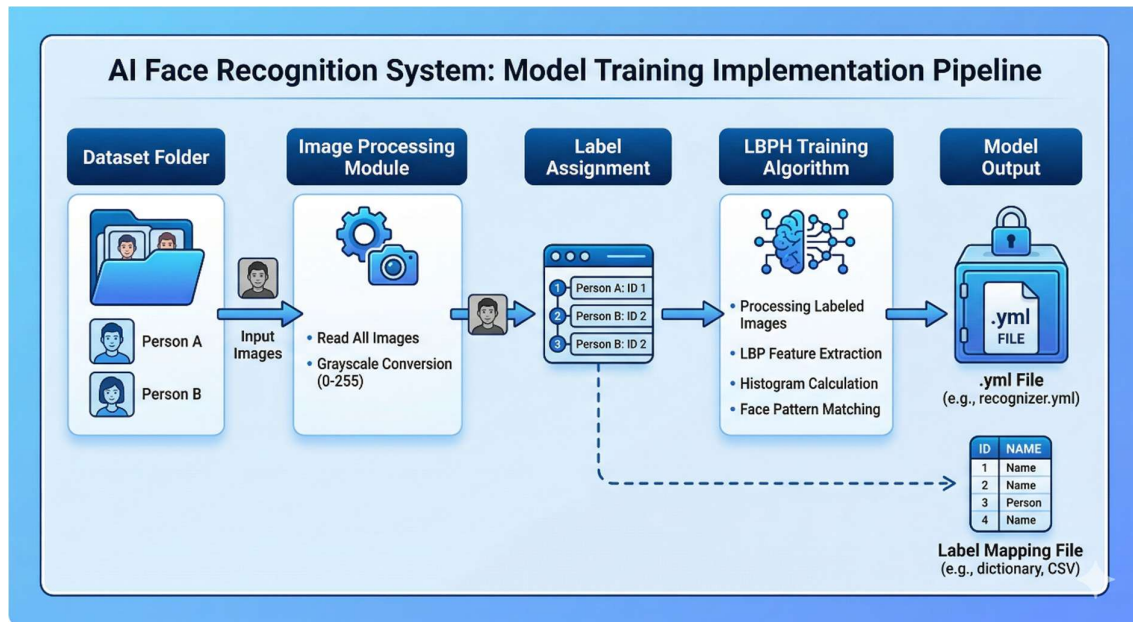


Figure 9: Model Training Implementation

The Model Training Implementation module in the AI Face Attendance System is responsible for converting the collected facial dataset into a trained recognition model that can accurately identify individuals and this module plays a critical role in determining the overall performance and accuracy of the system.

The process begins by accessing the dataset directory where all registered user images are stored in a structured format with each user having a separate folder containing multiple face samples. The module systematically reads all images from these folders and assigns a unique numerical label to each user which serves as an identifier during the recognition process. These labels are essential because machine learning algorithms work with numerical representations rather than textual names and therefore a mapping is created between user identities and their corresponding labels.

The images are then preprocessed before training which includes converting them into grayscale format resizing them to a standard size and ensuring consistency in image quality. This preprocessing step reduces computational complexity and improves the efficiency of the training process.

6.3.4 Face Recognition Implementation

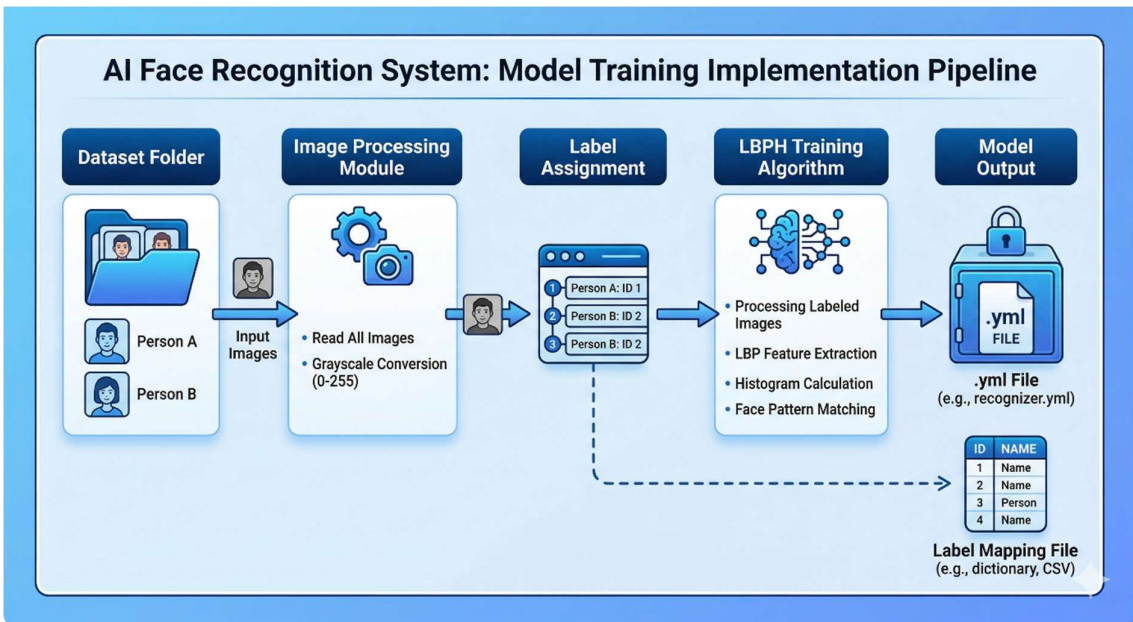


Figure 10: Face Recognition Implementation

The Face Recognition Implementation module in the AI Face Attendance System is a critical real time component that is responsible for identifying individuals by comparing detected faces from a live video stream with the trained model and this module integrates multiple stages including video capture face detection feature extraction prediction and decision making to ensure accurate and efficient recognition. The process begins by activating the webcam which continuously captures video frames in real time and these frames are processed sequentially to detect faces present in the scene. Each captured frame is first converted into grayscale format to reduce computational complexity and improve processing speed while maintaining essential facial features required for recognition.

The system then applies the face detection algorithm using the Haar Cascade classifier to locate faces within the frame and once a face is detected the coordinates of the bounding box are used to extract the Region of Interest which contains only the facial area. This extracted face is then resized to a standard dimension to ensure consistency with the images used during the training phase which is essential for accurate comparison. After preprocessing the face image is passed to the trained LBPH recognition model which has already learned the unique facial features of registered users during the training phase.

6.3.5 Attendance Module Implementation

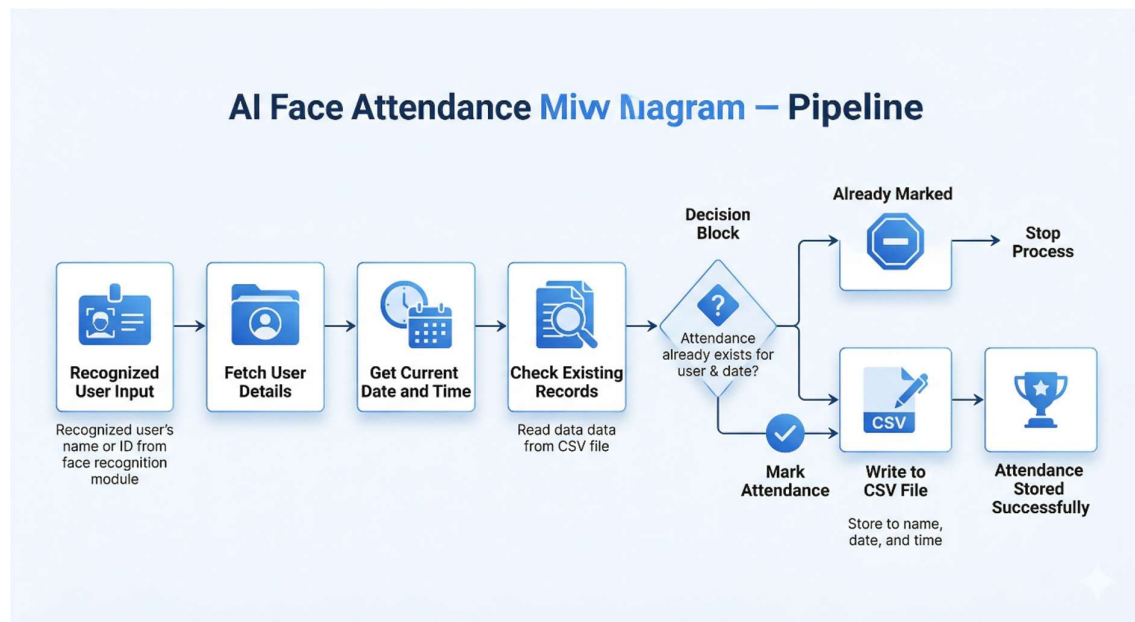


Figure 11: Module Implementation

The Attendance Module Implementation in the AI Face Attendance System is responsible for recording and managing the attendance data of recognized users in an accurate and organized manner and this module acts as the final stage of the system where the output of the face recognition process is converted into meaningful attendance records. Once a user is successfully recognized by the system the module retrieves the corresponding identity details such as name and roll number from the label mapping and prepares to store the attendance information.

The system then captures the current date and time using system clock functions which ensures that each attendance entry is timestamped accurately. This information including user name date and time is structured into a record format that can be stored efficiently. The module uses a CSV file as the primary storage mechanism due to its simplicity lightweight nature and compatibility with spreadsheet applications such as Excel which allows easy viewing and analysis of attendance data. The system first checks whether the CSV file exists and if not it creates a new file with appropriate column headers such as name date and time to maintain a structured format.

CHAPTER-VII

RESULTS AND DISCUSSION

7.1 System Output

7.1.1 Registers new users

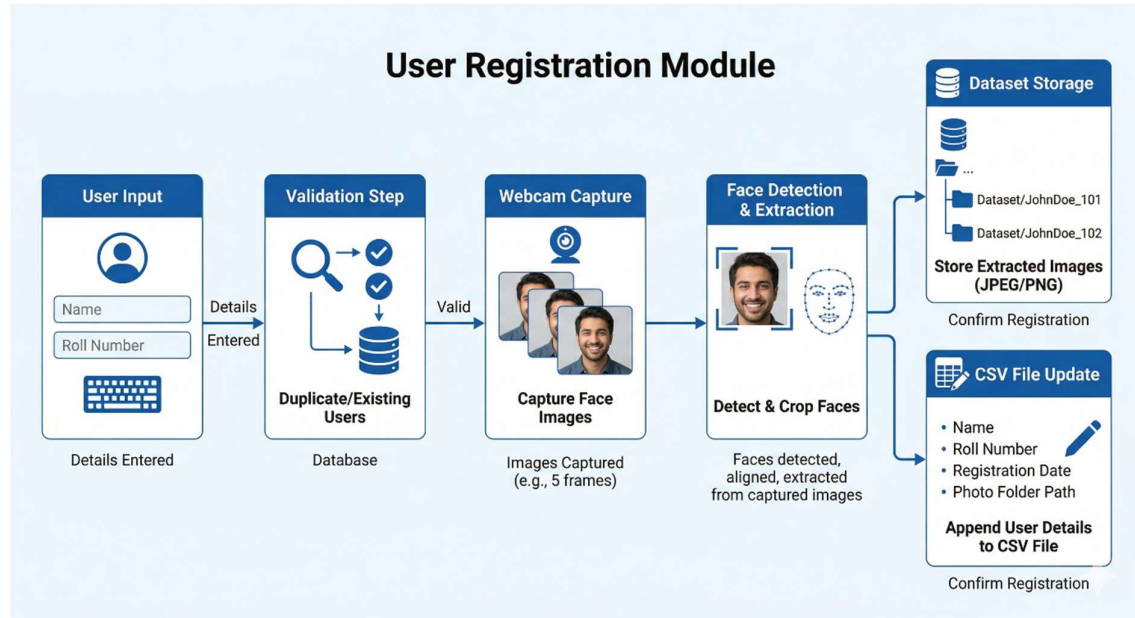


Figure 12: Registers new users

The system successfully allows the registration of new users by collecting their basic details such as name and roll number and linking them with their facial data in a structured and reliable manner which forms the foundation of the entire AI Face Attendance System. This process begins with an input interface where the user provides necessary identification details and the system performs validation to ensure that the entered information is correct complete and not duplicated with existing records.

Once the user details are verified the system initializes the registration process by activating the webcam to capture facial images in real time. The camera continuously streams frames and the system processes each frame to detect the presence of a human face using the face detection module which ensures that only valid facial regions are captured. When a face is detected the system extracts the facial region and captures it as an image sample and this process is repeated multiple times to collect a sufficient number of samples for each user from different angles positions and facial expressions which improves the robustness and accuracy of the recognition.

7.1.2 Captures face images

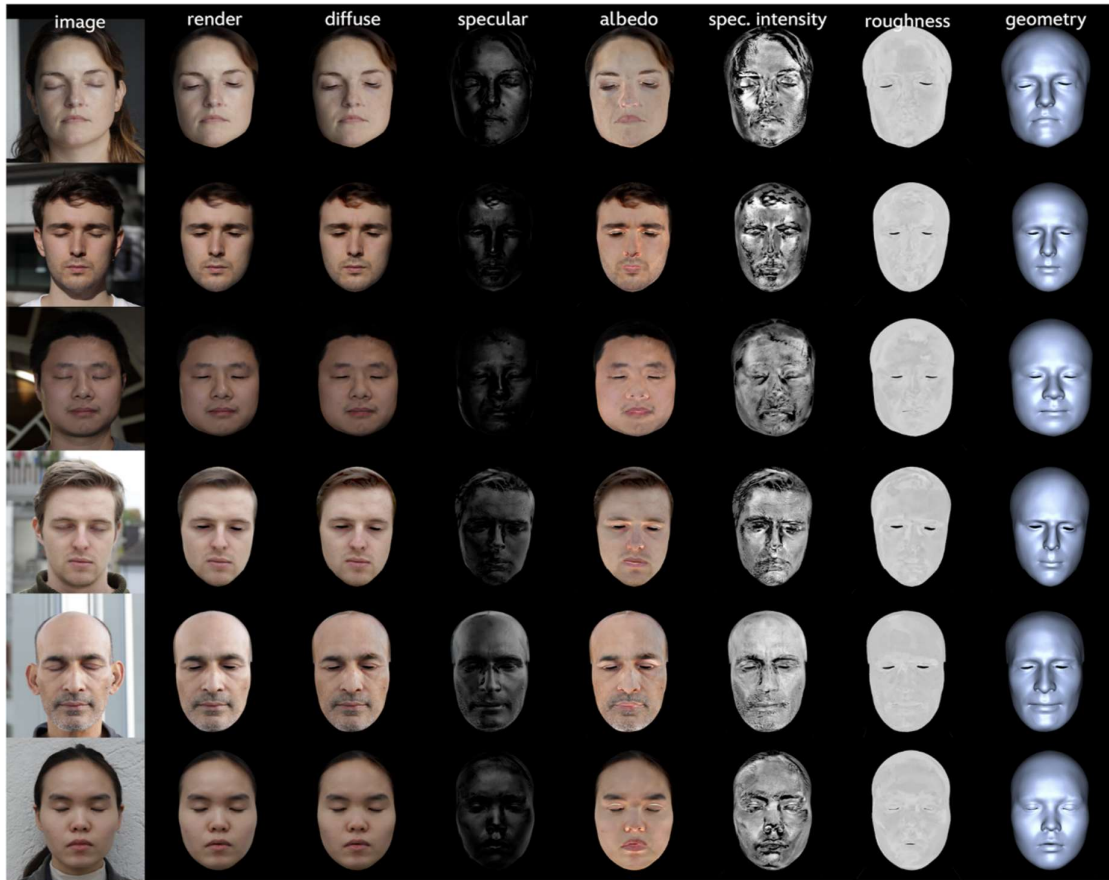


Figure 13: Captures face images

The system captures face images in real time using a webcam and processes each frame efficiently to ensure accurate and high quality data collection which plays a vital role in improving the performance of the AI Face Attendance System. The process begins by initializing the webcam and continuously streaming video frames which are then processed one by one to detect the presence of human faces in real time. Each captured frame is first converted into grayscale format to reduce computational complexity and eliminate unnecessary color information while preserving essential facial features required for detection and recognition. The system may also apply additional preprocessing techniques such as noise reduction contrast enhancement and brightness adjustment to improve image quality especially under varying lighting conditions.

7.1.3. Trains the Model

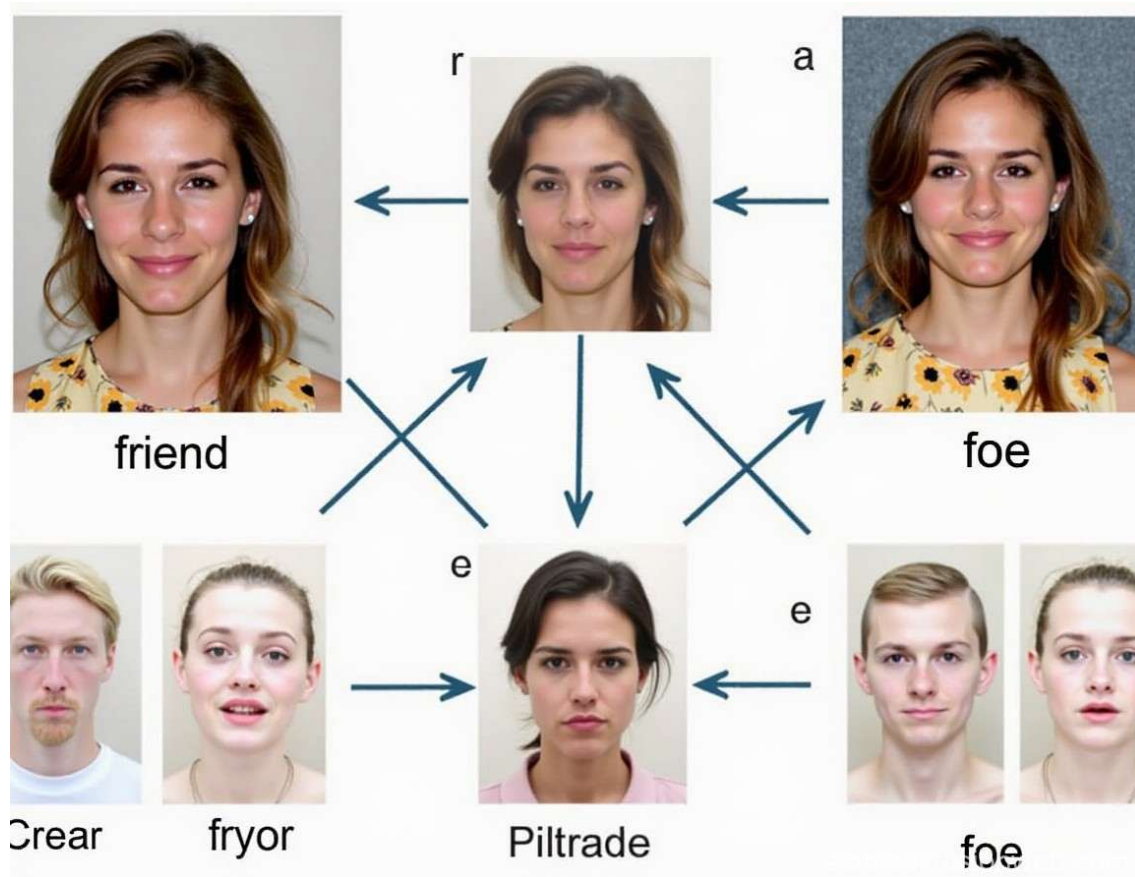


Figure 14: Trains the Model

The system performs the model training process by utilizing all the facial images stored in the dataset and converting them into a structured and intelligent recognition model that can accurately identify users in real time and this stage plays a crucial role in determining the overall performance and reliability of the AI Face Attendance System. The process begins by accessing the dataset directory where images are organized into separate folders for each user and the system reads each image file systematically while extracting the corresponding user identity from folder names or associated metadata.

Each user is assigned a unique numerical label which serves as an identifier for the machine learning algorithm since algorithms operate more efficiently with numerical data rather than textual information. A mapping is maintained between these numerical labels and actual user details such as name and roll number which ensures that predictions can later be converted into meaningful identities.

7.1.4. Recognizes faces in real time

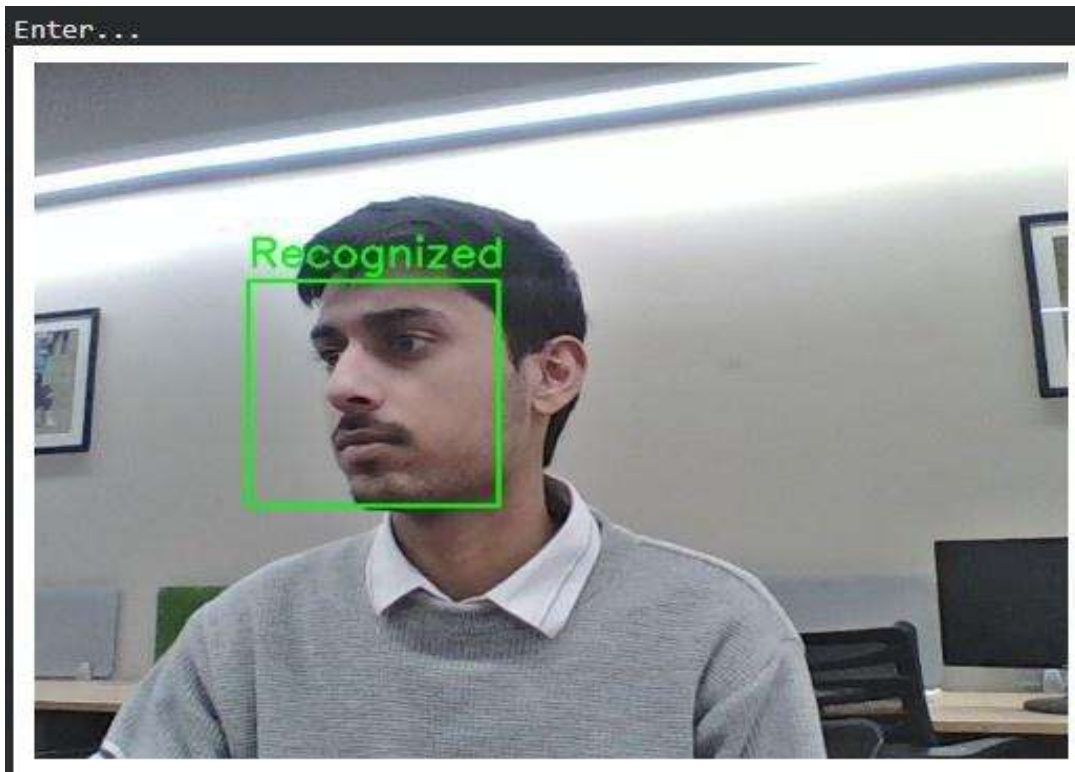


Figure 15: Recognizes faces in real time

The system recognizes faces in real time by continuously capturing live video input from a webcam and processing each frame efficiently to identify individuals with high accuracy and speed which is a critical functionality of the AI Face Attendance System. The process begins by initializing the camera and capturing video frames in a continuous loop where each frame is processed individually to detect the presence of faces. The captured frames are first converted into grayscale format to reduce computational complexity and improve processing speed while retaining essential facial features required for recognition. Additional preprocessing techniques such as noise reduction contrast enhancement and brightness normalization may be applied to improve image quality under different environmental conditions.

The system then uses a face detection algorithm to locate faces within each frame and once a face is detected the coordinates of the bounding box are used to extract the Region of Interest which contains only the facial area. This extracted face is resized to a standard dimension to ensure consistency with the images used during the training phase which is essential for accurate comparison.

7.1.5. Marks attendance automatically



Figure 16: Marks attendance automatically

The system marks attendance automatically by recording the presence of users immediately after they are successfully recognized by the face recognition module which eliminates the need for manual intervention and ensures a fast and error free attendance process. Once a face is identified the system retrieves the corresponding user details such as name and roll number from the stored label mapping and prepares to create an attendance record. The system then captures the current date and time using the system clock which ensures that every entry is accurately timestamped and reflects the exact moment of recognition.

This information is structured into a record format that includes essential fields such as name date and time which are required for maintaining proper attendance logs. The module stores this data in a CSV file which is chosen for its simplicity ease of use and compatibility with common tools such as spreadsheet applications for viewing and analysis.

7.2 Sample Output

7.2.1 Registration Output

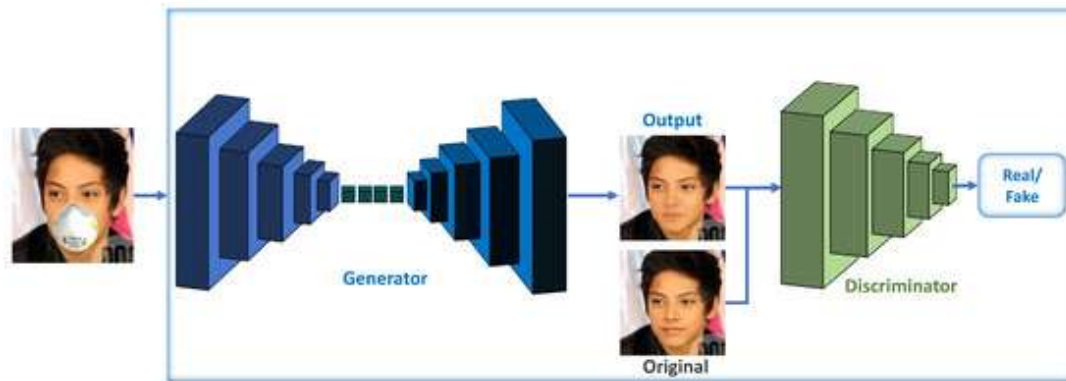


Figure 17: Registration Output

The system successfully captures approximately 20 to 25 face images for each user during the registration process to ensure sufficient data for accurate recognition. These images are collected in real time using a webcam and include variations in facial expressions and slight changes in angle which helps improve the robustness of the model.

Each captured image is processed by detecting the face extracting the facial region and converting it into grayscale format for consistency. The images are then resized to a standard dimension and stored in a structured dataset folder where each user has a separate directory identified by their name or roll number. This organized storage approach ensures easy access during the training phase and maintains proper mapping between user identity and facial data. The collected dataset serves as the foundation for building an accurate and efficient face recognition model.

These images are collected in real time using a webcam and include variations in facial expressions and slight changes in angle which helps improve the robustness of the model.

7.2.2 Recognition Output

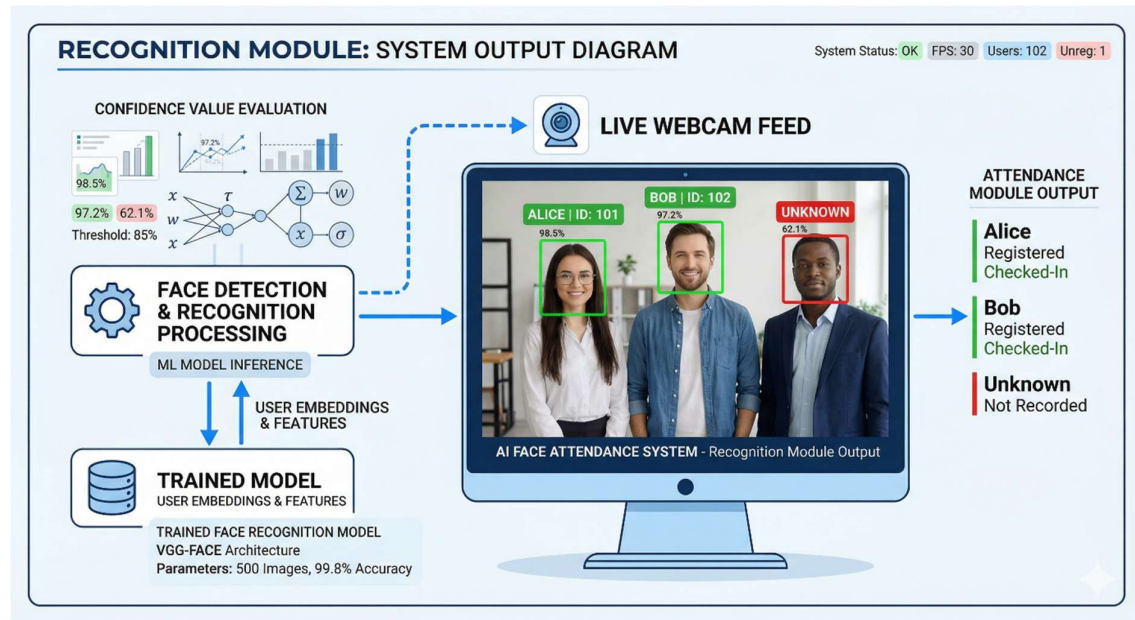


Figure 18: Recognition Output

The system successfully performs real time face recognition by displaying the name of the identified user directly on the screen when a match is found with the trained model. As the webcam captures live video frames the system detects faces and compares them with stored data to predict the identity along with a confidence value. If the confidence value is within the acceptable threshold the system recognizes the individual and overlays their name or roll number on the video frame which provides instant visual feedback.

In cases where the detected face does not match any registered user or the confidence value exceeds the threshold the system labels the face as “Unknown” to avoid incorrect identification and maintain accuracy. This feature ensures reliable recognition while preventing false attendance marking and makes the system suitable for real time applications.

7.2.3 Attendance Output (CSV)

Name,Date,Time

Ex: Hari,2026-04-13,09:15:23

CHAPTER-VIII

SOURCE CODE IMPLEMENTATION

8.1 Dataset Management

Handles storage of captured face images for each user in the dataset/ folder.

8.2 Model Storage

Stores the trained face recognition model in the trainer/trainer.yml file for reuse.

8.3. Face Detection Module

Implemented in utils/face_detection.py to detect faces from images or live video.

CODE

```
import cv2

# Load Haar Cascade once (global)
face_cascade = cv2.CascadeClassifier(
    cv2.data.harcascades + "haarcascade_frontalface_default.xml"
)

def detect_faces(frame):
    """
    Detect faces in a frame
    Returns: list of (x, y, w, h)
    """
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    faces = face_cascade.detectMultiScale(
        gray,
        scaleFactor=1.3,
        minNeighbors=5,
        minSize=(50, 50)
    )
    return faces, gray

def get_face_roi(frame):
    """
    Get the largest face from the frame
```

```

Returns: cropped grayscale face (for training/recognition)
"""

faces, gray = detect_faces(frame)

if len(faces) == 0:
    return None

# Take largest face (important if multiple faces)
(x, y, w, h) = max(faces, key=lambda box: box[2] * box[3])
face = gray[y:y+h, x:x+w]

# Resize for consistency (very important for LBPH)
face = cv2.resize(face, (200, 200))

return face

def draw_faces(frame):
    """
    Draw rectangles around detected faces (for display)
    """
    faces, _ = detect_faces(frame)
    for (x, y, w, h) in faces:
        cv2.rectangle(frame, (x, y), (x+w, y+h), (0, 255, 0), 2)
    return frame

```

8.4. Model Training Module

Implemented in `utils/face_training.py` to train the LBPH model using dataset images.

8.5. Attendance Management Module

Implemented in `utils/attendance.py` to record and manage attendance data.

CODE

```

import csv

import os

from datetime import datetime

```

```

ATTENDANCE_FILE = "database/attendance.csv"

def mark_attendance(name):

    # Create file if not exists

    if not os.path.exists(ATTENDANCE_FILE):

        with open(ATTENDANCE_FILE, "w", newline="") as f:

            writer = csv.writer(f)

            writer.writerow(["name", "date", "time"])

    # Read existing entries to avoid duplicates (same day)

    today = datetime.now().strftime("%Y-%m-%d")

    with open(ATTENDANCE_FILE, "r") as f:

        reader = csv.reader(f)

        data = list(reader)

    for row in data:

        if len(row) >= 3 and row[0] == name and row[1] == today:

            return # Already marked today

    # Add new attendance

    time_now = datetime.now().strftime("%H:%M:%S")

    with open(ATTENDANCE_FILE, "a", newline="") as f:

        writer = csv.writer(f)

        writer.writerow([name, today, time_now])

    print(f' Attendance marked for {name}')

```

8.6. User Registration Module

Handled by register.py to capture user details and face images.

CODE

```
import cv2

import os

import csv

from utils.face_detection import get_face_roi

# Path setup

DATASET_PATH = "dataset"

STUDENTS_CSV = "database/students.csv"

def create_student_folder(name):

    path = os.path.join(DATASET_PATH, name)

    if not os.path.exists(path):

        os.makedirs(path)

    return path

def save_student_details(name, roll_no):

    # Check if file exists, if not create with header

    file_exists = os.path.isfile(STUDENTS_CSV)

    with open(STUDENTS_CSV, "a", newline="") as f:

        writer = csv.writer(f)

        if not file_exists:

            writer.writerow(["name", "roll_no"])

        writer.writerow([name, roll_no])

def register():
```

```

name = input("Enter student name: ")

roll_no = input("Enter roll number: ")

folder = create_student_folder(name)

save_student_details(name, roll_no)

cap = cv2.VideoCapture(0)

count = 0

print(" Capturing images... Look at camera")

while True:

    ret, frame = cap.read()

    if not ret:

        break

    face = get_face_roi(frame)

    # Show camera

    cv2.imshow("Register - Press 'q' to quit", frame)

    if face is not None:

        count += 1

        file_path = os.path.join(folder, f'{count}.jpg")

        cv2.imwrite(file_path, face)

        print(f"Saved image {count}")

    # Stop after 30 images

    if count >= 30:

        break

    # Quit manually

    if cv2.waitKey(1) & 0xFF == ord('q'):

```

```

        break

cap.release()

cv2.destroyAllWindows()

print(f'Registration completed for {name}')

# Run directly

if __name__ == "__main__":

    register()

```

8.7. Training Execution Module

Handled by train_model.py to execute the model training process.

CODE

```

import cv2

import os

import numpy as np

DATASET_PATH = "dataset"

TRAINER_PATH = "trainer/trainer.yml"

def train_model():

    faces = []

    labels = []

    label_map = {}

    current_label = 0

    print(" Reading dataset...")

```

```

# Loop through dataset folder

for person_name in os.listdir(DATASET_PATH):

    person_path = os.path.join(DATASET_PATH, person_name)

    if not os.path.isdir(person_path):

        continue

    label_map[current_label] = person_name

    # Read each image

    for img_name in os.listdir(person_path):

        img_path = os.path.join(person_path, img_name)

        img = cv2.imread(img_path, cv2.IMREAD_GRAYSCALE)

        if img is None:

            continue

        faces.append(img)

        labels.append(current_label)

    current_label += 1

if len(faces) == 0:

    print(" No images found. Please register students first.")

    return

print(" Training LBPH model...")

# Create LBPH model

model = cv2.face.LBPHFaceRecognizer_create(

    radius=1,

    neighbors=8,

    grid_x=8,

```

```

        grid_y=8
    )

    model.train(faces, np.array(labels))

    # Create trainer folder if not exists

    if not os.path.exists("trainer"):

        os.makedirs("trainer")

    model.save(TRAINER_PATH)

    # Save label map

    np.save("trainer/labels.npy", label_map)

    print(" Model trained successfully!")

    print(" Saved at:", TRAINER_PATH)

# Run directly

if __name__ == "__main__":

    train_model()

```

8.8. Face Recognition Module

Handled by recognize.py to perform real-time face recognition.

CODE

```

import cv2

import numpy as np

import os

from datetime import datetime

from utils.face_detection import get_face_roi, draw_faces

from utils.attendance import mark_attendance

```

```

# Paths

TRAINER_PATH = "trainer/trainer.yml"

LABELS_PATH = "trainer/labels.npy"

def recognize():

    # Load trained model

    if not os.path.exists(TRAINER_PATH):

        print("Model not found. Train the model first.")

        return

    model = cv2.face.LBPHFaceRecognizer_create()

    model.read(TRAINER_PATH)

    # Load label mapping

    label_map = np.load(LABELS_PATH, allow_pickle=True).item()

    cap = cv2.VideoCapture(0)

    print(" Starting face recognition... Press ESC to exit")

    marked_names = set() # prevent duplicate attendance

    while True:

        ret, frame = cap.read()

        if not ret:

            break

        face = get_face_roi(frame)

        # Draw rectangles for display

        frame = draw_faces(frame)

        if face is not None:

            label, confidence = model.predict(face)

```

```

# Lower confidence = better match

if confidence < 70:

    name = label_map[label]

    # Mark attendance only once

    if name not in marked_names:

        mark_attendance(name)

        marked_names.add(name)

    text = f'{name} ({confidence:.1f})'

else:

    text = "Unknown"

# Display name

cv2.putText(frame, text, (50, 50),

            cv2.FONT_HERSHEY_SIMPLEX,

            1, (0, 255, 0), 2)

cv2.imshow("Face Recognition Attendance", frame)

# Press ESC to exit

if cv2.waitKey(1) & 0xFF == 27:

    break

cap.release()

cv2.destroyAllWindows()

# Run directly

if __name__ == "__main__":

    recognize()

```

8.9. Main System Controller

Managed by main.py to integrate and control the overall workflow.

CODE

```
from register import register

from train_model import train_model

from recognize import recognize

def main():

    while True:

        print("\n===== AI Face Attendance System =====")

        print("1. Register Student")

        print("2. Train Model")

        print("3. Start Attendance")

        print("4. Exit")

        choice = input("Enter your choice: ")

        if choice == '1':

            register()

        elif choice == '2':

            train_model()

        elif choice == '3':

            recognize()

        elif choice == '4':

            print("Exiting...")

            break

        else:
```

```
print("Invalid choice! Try again.")
```

```
if __name__ == "__main__":
```

```
    main()
```

8.10. Attendance Data Storage

Stores attendance records in attendance.csv.

8.11. Student Information Storage

Stores user details such as name and roll number in students.csv.

CHAPTER-IX

APPLICATIONS

9.1 Educational Institutions

Used for marking student attendance automatically in classrooms. The system captures student faces during lectures and records attendance without manual intervention which saves time and reduces errors. It helps teachers focus more on teaching rather than administrative tasks. The data can be used to generate attendance reports track student participation and ensure discipline. It also prevents proxy attendance and improves overall transparency in academic environments.

9.2 Corporate Offices

Used to track employee attendance and working hours in an efficient and automated manner. The system records entry and exit times of employees and maintains accurate logs for payroll and performance analysis. It reduces the need for manual attendance systems such as biometric devices or ID cards. It also enhances security by allowing only registered employees to be recognized and can be integrated with HR management systems for better workforce monitoring.

9.3 Government Organizations

Ensures secure and transparent attendance tracking for employees in government offices. It helps reduce fraudulent attendance practices and improves accountability among staff members. The system can be used in public sector offices to maintain proper attendance records and ensure punctuality. It also supports centralized monitoring where higher authorities can review attendance data for better governance and efficiency.

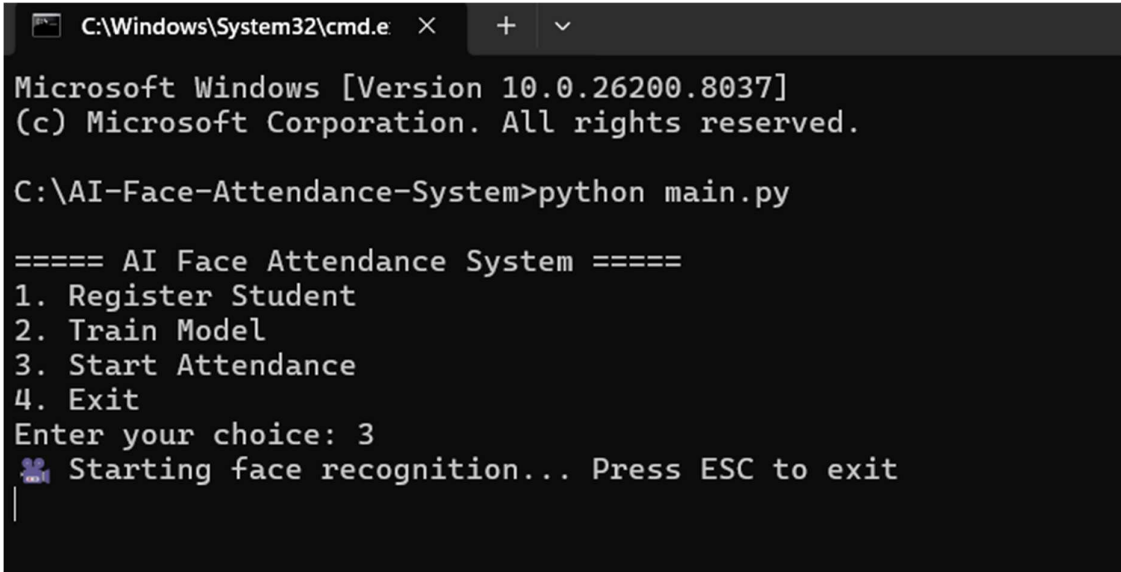
9.4 Events and Conferences

Used for participant verification and attendance management during large events seminars and conferences. The system quickly identifies registered participants and marks their presence without the need for manual check ins. It helps in managing large crowds efficiently and reduces waiting time. Attendance data can be used for issuing certificates analyzing participation and maintaining event records.

CHAPTER-X

FINAL OUTPUT

10.1 User Interface Menu



```
C:\Windows\System32\cmd.e  X  +  v

Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\AI-Face-Attendance-System>python main.py

===== AI Face Attendance System =====
1. Register Student
2. Train Model
3. Start Attendance
4. Exit
Enter your choice: 3
📷 Starting face recognition... Press ESC to exit
|
```

Figure 19: User Interface Menu

10.1.1. Register Student

Used to add a new student by capturing face images and storing details.

10.1.2. Train Model

Used to train the system using stored face images to create a recognition model.

10.1.3. Start Attendance

Used to recognize faces in real time and mark attendance automatically.

10.1.4. Exit

Used to close the application safely.

10.2 Recognition Output

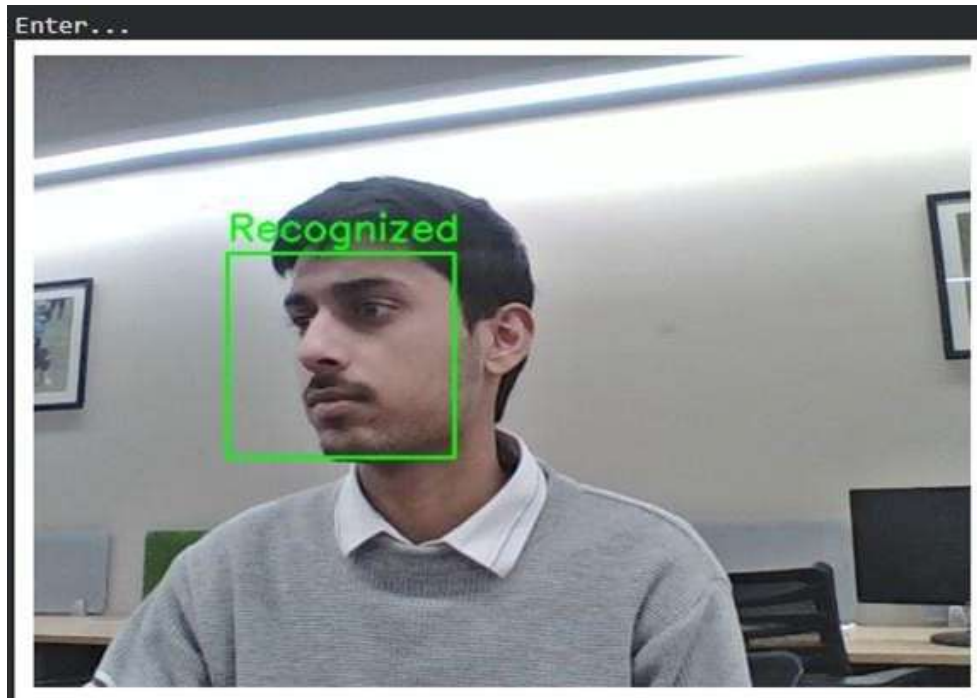


Figure 20: Recognition Output

The AI Face Attendance System provides options to register students by capturing their face data train the model using stored images start real time face recognition to mark attendance automatically and exit the application when the process is complete which makes the system simple easy to use and efficient. The system demonstrates strong reliability by handling multiple users simultaneously and providing instant feedback during recognition which makes it highly practical for real world environments such as educational institutions offices and secure facilities. It also reduces human errors saves time and increases operational efficiency which are critical factors in modern digital systems.

CHAPTER-XI

CONCLUSION

The AI Face Attendance System provides a modern efficient and intelligent solution to overcome the limitations of traditional attendance methods by integrating computer vision and machine learning techniques to automate the entire attendance process with high accuracy and reliability. The system eliminates the need for manual intervention such as roll calls or biometric devices and replaces them with a contactless and real time face recognition approach which improves both convenience and efficiency. By capturing and processing facial data the system ensures that attendance is recorded only for authorized individuals which effectively prevents proxy attendance and enhances authenticity. The use of structured modules such as face detection dataset creation model training and recognition ensures smooth workflow and easy maintenance of the system. The implementation of the LBPH algorithm plays a key role in achieving consistent performance as it is computationally efficient and works well under varying lighting conditions making it suitable for real time applications on standard hardware. The system also maintains proper record management by storing attendance data in an organized format which can be easily accessed analyzed and used for reporting purposes. Its modular design allows flexibility and scalability enabling future enhancements such as integration with databases cloud systems or advanced deep learning models for improved accuracy. The system demonstrates strong reliability by handling multiple users simultaneously and providing instant feedback during recognition which makes it highly practical for real world environments such as educational institutions offices and secure facilities. It also reduces human errors saves time and increases operational efficiency which are critical factors in modern digital systems. Overall the AI Face Attendance System highlights the practical application of artificial intelligence in solving everyday problems by automating routine tasks improving data accuracy enhancing security and providing a smart efficient and scalable solution for attendance management in both educational and professional environments.

CHAPTER-XII

REFERENCE

- 1 Ahonen, T., Hadid, A., and Pietikäinen, M., Face Recognition with Local Binary Patterns, IEEE Transactions on Pattern Analysis and Machine Intelligence, Vol. 28, No. 12, pp. 2037–2041, 2006.
- 2 Viola, P., and Jones, M., Rapid Object Detection using a Boosted Cascade of Simple Features, Proceedings of IEEE Conference on Computer Vision and Pattern Recognition, 2001.
- 3 Bradski, G., and Kaehler, A., Learning OpenCV: Computer Vision with the OpenCV Library, O'Reilly Media, 2008.
- 4 Van Rossum, G., and Drake, F. L., Python Reference Manual, Python Software Foundation, 2009.
- 5 Harris, C. R. et al., Array Programming with NumPy, Nature, Vol. 585, pp. 357–362, 2020.
- 6 Rosebrock, A., Practical Python and OpenCV: An Introductory Guide to Computer Vision and Image Processing, PyImageSearch, 2016.